Laws of Indices

Worked Solutions with TikZ Diagrams

Wisest Maths — Year 1 A-Level, Algebra (ref `a4`)

This document contains all 50 *Laws of Indices* questions, each with a fully worked solution and a TikZ diagram matched to the law in play:

- **Law cards** — the boxed general rule with the exponent arithmetic — for the multiplication, division, power-of-a-power, power-of-a-product, negative-power and zero-power laws;
- **Power/root split** diagrams for fractional indices (denominator = root, numerator = power);
- **Solution-flow** diagrams (one law per arrow) for multi-step numeric chains and for solving $a^x = k$.

Contents

Laws of Indices 01 `(a4-001)`

Difficulty: Foundation | Marks: 1

Simplify, leaving your answer as a power: $5^3 \times 5^4$

Worked solution.

Step 1: When multiplying powers with the same base, add the exponents.

$$5^3 \times 5^4 = 5^{3+4}$$

The multiplication law of indices says $a^m \times a^n = a^{m+n}$.

Step 2: Add the exponents together.

$$5^7$$

$3 + 4 = 7$.

Final answer: 5^7

Diagram.

$$\boxed{a^m \times a^n = a^{m+n}}$$

$$5^3 \times 5^4 \xrightarrow[\text{same base: add the powers, } 3+4=7]{} = 5^7$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 02 `(a4-002)`

Difficulty: Foundation | *Marks: 1*

Simplify, leaving your answer as a power: $8^5 \div 8^2$

Worked solution.

Step 1: When dividing powers with the same base, subtract the exponents.

$$8^5 \div 8^2 = 8^{5-2}$$

The division law of indices says $a^m \div a^n = a^{m-n}$.

Step 2: Subtract the exponents.

$$8^3$$

$5 - 2 = 3$.

Final answer: 8^3

Diagram.

$$\boxed{a^m \div a^n = a^{m-n}}$$

$$8^5 \div 8^2 \xrightarrow[\text{same base: subtract the powers, } 5-2=3]{} = 8^3$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 03 `(a4-003)`

Difficulty: Foundation | *Marks: 1*

Simplify, leaving your answer as a power: $(6^3)^2$

Worked solution.

Step 1: When raising a power to another power, multiply the exponents.

$$(6^3)^2 = 6^{3\times 2}$$

The power-of-a-power law says $(a^m)^n = a^{mn}$.

Step 2: Multiply the exponents together.

$$6^6$$

$3 \times 2 = 6$.

Final answer: 6^6

Diagram.

$$\boxed{(a^m)^n = a^{mn}}$$

$$(6^3)^2 \xrightarrow[\text{power of a power: multiply, } 3 \times 2 = 6]{} = \; 6^6$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 04 `(a4-004)`

Difficulty: Foundation | *Marks: 2*

Simplify, leaving your answer as a single power of y: $y^{-1} \times y^2 \times y^3$

Worked solution.

Step 1: Add all the exponents together since the bases are the same and the terms are multiplied.

$$y^{-1+2+3}$$

The multiplication law applies to any number of terms: $a^m \times a^n \times a^p = a^{m+n+p}$.

Step 2: Simplify the sum of the exponents.

$$y^4$$

$-1 + 2 + 3 = 4$.

Final answer: y^4

Diagram.

$$\boxed{a^m \times a^n = a^{m+n}}$$

$$y^{-1} \times y^2 \times y^3 \xrightarrow[\text{add the powers: } -1+2+3=4]{} = \; y^4$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 05 `(a4-005)`

Difficulty: Foundation | *Marks: 1*

Simplify, leaving your answer as a power: $\frac{6^{11}}{6^6}$

Worked solution.

Step 1: Writing the fraction as a division and subtracting exponents.

$$\frac{6^{11}}{6^6} = 6^{11-6}$$

A fraction with the same base on top and bottom is the same as dividing, so subtract the exponents.

Step 2: Simplify the exponent.

$$6^5$$

$11 - 6 = 5$.

Final answer: 6^5

Diagram.

$$\boxed{\frac{a^m}{a^n} = a^{m-n}}$$

$$\frac{6^{11}}{6^6} \xrightarrow[\text{subtract the powers: } 11-6=5]{} = 6^5$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 06 `(a4-006)`

Difficulty: Foundation | *Marks: 1*

Simplify, leaving your answer as a single power of r: $\frac{r^2}{r^6}$

Worked solution.

Step 1: Subtract the exponents using the division law.

$$r^{2-6}$$

Subtract the bottom exponent from the top exponent.

Step 2: Simplify the exponent.

$$r^{-4}$$

$2 - 6 = -4$. A negative exponent is perfectly valid here.

Final answer: r^{-4}

Diagram.

$$\boxed{\frac{a^m}{a^n} = a^{m-n}}$$

$$\frac{r^2}{r^6} \xrightarrow[\text{subtract: } 2-6=-4]{} = r^{-4}$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 07 `(a4-007)`

Difficulty: Foundation | *Marks: 1*

Simplify: $(k^{-2})^5$

Worked solution.

Step 1: Multiply the exponents using the power-of-a-power law.

$$k^{-2\times5}$$

$(a^m)^n = a^{mn}$, even when m is negative.

Step 2: Evaluate the product of the exponents.

$$k^{-10}$$

$-2 \times 5 = -10$.

Final answer: k^{-10}

Diagram.

$$\boxed{(a^m)^n = a^{mn}}$$

$$(k^{-2})^5 \xrightarrow[\text{multiply: } -2 \times 5 = -10]{} = \; k^{-10}$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 08 `(a4-008)`

Difficulty: Foundation | Marks: 2

Evaluate: $4^{\frac{1}{2}} \times 4^{\frac{1}{2}}$

Worked solution.

Step 1: Add the fractional exponents since the bases are the same.

$$4^{\frac{1}{2}+\frac{1}{2}} = 4^1$$

$\frac{1}{2} + \frac{1}{2} = 1$.

Step 2: Evaluate.

$$4$$

Any number to the power 1 is itself.

Final answer: 4

Diagram.

$$\boxed{a^m \times a^n = a^{m+n}}$$

$$4^{\frac{1}{2}} \times 4^{\frac{1}{2}} \xrightarrow[\text{add: } \frac{1}{2} + \frac{1}{2} = 1, \text{ so } 4^1 = 4]{} = \; 4$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 09 `(a4-009)`

Difficulty: Foundation | *Marks: 2*

Evaluate: $3^4 \div 3^1$

Worked solution.

Step 1: Subtract the exponents using the division law.

$$3^{4-1} = 3^3$$

$a^m \div a^n = a^{m-n}$.

Step 2: Evaluate 3^3.

$$3 \times 3 \times 3 = 27$$

Multiply three copies of 3 together.

Final answer: 27

Diagram.

$$\boxed{a^m \div a^n = a^{m-n}}$$

$$3^4 \div 3^1 \xrightarrow[\text{subtract: } 4-1=3\text{, so } 3^3=27]{\qquad\qquad\qquad} = \ 27$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 10 `(a4-010)`

Difficulty: Foundation | *Marks: 2*

Evaluate: $(2^3)^2 \div 2^4$

Worked solution.

Step 1: Apply the power-of-a-power law to simplify $(2^3)^2$.

$$2^{3\times 2} \div 2^4 = 2^6 \div 2^4$$

Multiply the exponents inside the bracket first.

Step 2: Now apply the division law.

$$2^{6-4} = 2^2$$

Subtract the exponents since the bases are the same.

Step 3: Evaluate 2^2.

$$4$$

$2 \times 2 = 4$.

Final answer: 4

Diagram.

$$(2^3)^2 \div 2^4 \xrightarrow{\text{power of a power}} 2^6 \div 2^4 \xrightarrow{\text{subtract powers}} 2^2 = 4$$

Solution flow: each arrow applies one index law.

Laws of Indices 11 (a4-011)

Difficulty: Foundation | Marks: 1

Evaluate: 7^0

Worked solution.

Apply the zero exponent rule.

$$7^0 = 1$$

Any non-zero number raised to the power 0 equals 1. This follows from the division law: $a^n \div a^n = a^{n-n} = a^0 = 1$.

Final answer: 1

Diagram.

$$\boxed{a^0 = 1}$$

$$7^0 \xrightarrow[\text{any non-zero base to the power 0 equals 1}]{} = \ 1$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 12 (a4-012)

Difficulty: Foundation | Marks: 1

Evaluate: $\left(\frac{3}{5}\right)^0$

Worked solution.

Apply the zero exponent rule.

$$\left(\frac{3}{5}\right)^0 = 1$$

Anything (except zero) raised to the power 0 is 1 — it does not matter whether it is a whole number or a fraction.

Final answer: 1

Diagram.

$$\boxed{a^0 = 1}$$

$$\left(\frac{3}{5}\right)^0 \xrightarrow[\text{any non-zero base to the power 0 equals 1}]{} = \ 1$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 13 `(a4-013)`

Difficulty: Foundation | *Marks: 1*

Evaluate: $9^{\frac{1}{2}}$

Worked solution.

Step 1: Recognise that a power of $\frac{1}{2}$ means the square root.

$$9^{\frac{1}{2}} = \sqrt{9}$$

$a^{\frac{1}{2}} = \sqrt{a}$.

Step 2: Evaluate the square root.

$$3$$

$3 \times 3 = 9$, so $\sqrt{9} = 3$.

Final answer: 3

Diagram.

numerator $= 1$ (power)

$$9^{\frac{1}{2}} \quad = \quad \sqrt{9} \quad = \quad 3$$

denominator $= 2$ (root)

Fractional index: the *denominator* gives the root, the *numerator* gives the power.

Laws of Indices 14 `(a4-014)`

Difficulty: Foundation | *Marks: 1*

Evaluate: $8^{\frac{1}{3}}$

Worked solution.

Step 1: Recognise that a power of $\frac{1}{3}$ means the cube root.

$$8^{\frac{1}{3}} = \sqrt[3]{8}$$

$a^{\frac{1}{3}} = \sqrt[3]{a}$.

Step 2: Evaluate the cube root.

$$2$$

$2 \times 2 \times 2 = 8$, so the cube root of 8 is 2.

Final answer: 2

Diagram.

numerator $= 1$ (power)

$$8^{\frac{1}{3}} \quad = \quad \sqrt[3]{8} \quad = \quad 2$$

denominator $= 3$ (root)

Fractional index: the *denominator* gives the root, the *numerator* gives the power.

Laws of Indices 15 `(a4-015)`

Difficulty: Foundation | Marks: 1

Evaluate: $27^{\frac{1}{3}}$

Worked solution.

Step 1: The exponent $\frac{1}{3}$ means take the cube root.

$$27^{\frac{1}{3}} = \sqrt[3]{27}$$

Ask yourself: what number multiplied by itself three times gives 27?

Step 2: Evaluate.

$$3$$

$3 \times 3 \times 3 = 27$.

Final answer: 3

Diagram.

numerator $= 1$ (power)

$$27^{\frac{1}{3}} \quad = \quad \sqrt[3]{27} \quad = \quad 3$$

denominator $= 3$ (root)

Fractional index: the *denominator* gives the root, the *numerator* gives the power.

Laws of Indices 16 `(a4-016)`

Difficulty: Foundation | Marks: 2

Evaluate: $16^{\frac{3}{4}}$

Worked solution.

Step 1: Split the fractional exponent: root first, then power.

$$16^{\frac{3}{4}} = \left(16^{\frac{1}{4}}\right)^3$$

The denominator 4 tells you to take the fourth root, and the numerator 3 tells you to cube the result.

Step 2: Evaluate the fourth root of 16.

$$16^{\frac{1}{4}} = 2$$

$2 \times 2 \times 2 \times 2 = 16$.

Step 3: Cube the result.

$$2^3 = 8$$

$2 \times 2 \times 2 = 8$.

Final answer: 8

Diagram.

numerator = 3 (power)

$$16^{\frac{3}{4}} \quad = \quad \left(\sqrt[4]{16}\right)^3 = 2^3 \quad = \; 8$$

denominator = 4 (root)

Fractional index: the *denominator* gives the root, the *numerator* gives the power.

Laws of Indices 17 (a4-017)

Difficulty: Foundation | Marks: 2

Evaluate: $25^{\frac{3}{2}}$

Worked solution.

Step 1: Split the fractional exponent: root first, then power.

$$25^{\frac{3}{2}} = \left(25^{\frac{1}{2}}\right)^3$$

The denominator 2 means square root; the numerator 3 means cube.

Step 2: Take the square root of 25.

$$25^{\frac{1}{2}} = 5$$

$5 \times 5 = 25$.

Step 3: Cube the result.

$$5^3 = 125$$

$5 \times 5 \times 5 = 125$.

Final answer: 125

Diagram.

numerator = 3 (power)

$$25^{\frac{3}{2}} \quad = \quad \left(\sqrt{25}\right)^3 = 5^3 \quad = \; 125$$

denominator = 2 (root)

Fractional index: the *denominator* gives the root, the *numerator* gives the power.

Laws of Indices 18 (a4-018)

Difficulty: Foundation | Marks: 2

Evaluate: $32^{\frac{2}{5}}$

Worked solution.

Step 1: Split the fractional exponent: fifth root first, then square.

$$32^{\frac{2}{5}} = \left(32^{\frac{1}{5}}\right)^2$$

The denominator 5 means take the fifth root; the numerator 2 means square.

Step 2: Evaluate the fifth root of 32.

$$32^{\frac{1}{5}} = 2$$

$2^5 = 32$.

Step 3: Square the result.

$$2^2 = 4$$

$2 \times 2 = 4$.

Final answer: 4

Diagram.

numerator $= 2$ (power)

$$32^{\frac{2}{5}} \quad = \quad \left(\sqrt[5]{32}\right)^2 = 2^2 \quad = \quad 4$$

denominator $= 5$ (root)

Fractional index: the *denominator* gives the root, the *numerator* gives the power.

Laws of Indices 19 (a4-019)

Difficulty: Foundation | Marks: 2

Evaluate: $4^{-\frac{1}{2}}$

Worked solution.

Step 1: Deal with the negative exponent by taking the reciprocal.

$$4^{-\frac{1}{2}} = \frac{1}{4^{\frac{1}{2}}}$$

A negative exponent means "one over": $a^{-n} = \frac{1}{a^n}$.

Step 2: Evaluate the square root of 4.

$$4^{\frac{1}{2}} = 2$$

$\sqrt{4} = 2$.

Step 3: Write the final answer.

$$\frac{1}{2}$$

Substituting back gives $\frac{1}{2}$.

Final answer: $\frac{1}{2}$

Diagram.

$$\boxed{a^{-n} = \frac{1}{a^n}}$$

$$4^{-\frac{1}{2}} \xrightarrow[\text{negative power} \Rightarrow \text{reciprocal: } \frac{1}{4^{1/2}} = \frac{1}{2}]{} = \frac{1}{2}$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 20 `(a4-020)`

Difficulty: Foundation | Marks: 3

Evaluate: $27^{-\frac{2}{3}}$

Worked solution.

Step 1: Deal with the negative sign by taking the reciprocal.

$$27^{-\frac{2}{3}} = \frac{1}{27^{\frac{2}{3}}}$$

Flip the base to make the exponent positive.

Step 2: Split the fractional exponent: cube root first, then square.

$$27^{\frac{2}{3}} = \left(27^{\frac{1}{3}}\right)^2 = 3^2 = 9$$

The cube root of 27 is 3, then $3^2 = 9$.

Step 3: Write the final answer.

$$\frac{1}{9}$$

So $27^{-\frac{2}{3}} = \frac{1}{9}$.

Final answer: $\frac{1}{9}$

Diagram.

$$\boxed{a^{-\frac{p}{q}} = \frac{1}{(\sqrt[q]{a})^p}}$$

$$27^{-\frac{2}{3}} \xrightarrow[\text{reciprocal of } 27^{2/3} = (\sqrt[3]{27})^2 = 9]{} = \frac{1}{9}$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 21 `(a4-021)`

Difficulty: Foundation | Marks: 1

Express $\frac{1}{m}$ as a power of m.

Worked solution.

"One over" a base is the same as that base raised to a negative power.

$$\frac{1}{m} = m^{-1}$$

By definition, $a^{-1} = \frac{1}{a}$.

Final answer: m^{-1}

Diagram.

$$\boxed{\frac{1}{a^n} = a^{-n}}$$

$$\frac{1}{m} \xrightarrow[\text{a denominator becomes a negative power}]{} = m^{-1}$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 22 `(a4-022)`

Difficulty: Foundation | Marks: 1

Express $\sqrt[3]{n}$ as a power of n.

Worked solution.

The cube root can be written as a fractional exponent.

$$\sqrt[3]{n} = n^{\frac{1}{3}}$$

The qth root of a number is the same as raising it to the power $\frac{1}{q}$.

Final answer: $n^{\frac{1}{3}}$

Diagram.

numerator $= 1$ (power)

$$\sqrt[3]{n} \quad = \quad n^{\frac{1}{3}} \quad = n^{\frac{1}{3}}$$

denominator $= 3$ (root)

Fractional index: the *denominator* gives the root, the *numerator* gives the power.

Laws of Indices 23 `(a4-023)`

Difficulty: Foundation | Marks: 2

Express $\frac{1}{\sqrt{t}}$ as a single power of t.

Worked solution.

Step 1: Rewrite the square root as a fractional exponent.

$$\frac{1}{\sqrt{t}} = \frac{1}{t^{\frac{1}{2}}}$$

$\sqrt{t} = t^{\frac{1}{2}}$.

Step 2: Use the negative exponent rule to bring the base to the numerator.

$$t^{-\frac{1}{2}}$$

$\frac{1}{a^n} = a^{-n}$.

Final answer: $t^{-\frac{1}{2}}$

Diagram.

$$\boxed{\frac{1}{\sqrt[q]{a}} = a^{-\frac{1}{q}}}$$

$$\frac{1}{\sqrt{t}} \xrightarrow[\sqrt{t} = t^{1/2}\text{, then reciprocal gives } t^{-1/2}]{} = t^{-\frac{1}{2}}$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 24 (a4-024)

Difficulty: Foundation | Marks: 2

Express $\left(\frac{1}{\sqrt[3]{x}}\right)^2$ as a single power of x.

Worked solution.

Step 1: Rewrite the expression using index notation.

$$\left(\frac{1}{x^{\frac{1}{3}}}\right)^2 = \left(x^{-\frac{1}{3}}\right)^2$$

First convert the cube root, then move it to the numerator with a negative exponent.

Step 2: Apply the power-of-a-power law.

$$x^{-\frac{1}{3}\times 2} = x^{-\frac{2}{3}}$$

Multiply the exponents: $-\frac{1}{3} \times 2 = -\frac{2}{3}$.

Final answer: $x^{-\frac{2}{3}}$

Diagram.

$$\left(\frac{1}{\sqrt[3]{x}}\right)^2 \xrightarrow{\text{rewrite as a power}} \left(x^{-\frac{1}{3}}\right)^2 \xrightarrow{\text{multiply powers}} x^{-\frac{2}{3}}$$

Solution flow: each arrow applies one index law.

Laws of Indices 25 (a4-025)

Difficulty: Foundation | Marks: 1

Simplify: $(z^4)^{\frac{1}{2}}$

Worked solution.

Multiply the exponents using the power-of-a-power law.

$$z^{4\times\frac{1}{2}} = z^2$$

$4 \times \frac{1}{2} = 2$. Taking the square root of z^4 gives z^2.

Final answer: z^2

Diagram.

$$\boxed{(a^m)^n = a^{mn}}$$

$$(z^4)^{\frac{1}{2}} \xrightarrow[\text{multiply: } 4 \times \frac{1}{2} = 2]{} = z^2$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 26 (a4-026)

Difficulty: Foundation / Marks: 2

Simplify: $(8^4)^{-\frac{1}{2}}$

Worked solution.

Step 1: Multiply the exponents.

$$8^{4\times(-\frac{1}{2})} = 8^{-2}$$

$4 \times (-\frac{1}{2}) = -2$.

Step 2: Deal with the negative exponent.

$$8^{-2} = \frac{1}{8^2} = \frac{1}{64}$$

$8^2 = 64$, so the answer is $\frac{1}{64}$.

Final answer: $\frac{1}{64}$

Diagram.

$$\boxed{(a^m)^n = a^{mn}}$$

$$(8^4)^{-\frac{1}{2}} \xrightarrow[\text{multiply: } 4 \times (-\frac{1}{2}) = -2, \text{ so } 8^{-2} = \frac{1}{64}]{} = \frac{1}{64}$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 27 (a4-027)

Difficulty: Foundation / Marks: 2

Evaluate: $\frac{2^3 \times 2}{2^5}$

Worked solution.

Step 1: Simplify the numerator using the multiplication law.

$$\frac{2^{3+1}}{2^5} = \frac{2^4}{2^5}$$

Remember $2 = 2^1$, so $2^3 \times 2^1 = 2^4$.

Step 2: Apply the division law.

$$2^{4-5} = 2^{-1}$$

Subtract the exponents.

Step 3: Evaluate.

$$2^{-1} = \frac{1}{2}$$

A power of -1 means the reciprocal.

Final answer: $\frac{1}{2}$

Diagram.

$$\frac{2^3 \times 2}{2^5} \xrightarrow{\text{add (top)}} \frac{2^4}{2^5} \xrightarrow{\text{subtract}} 2^{-1} = \frac{1}{2}$$

Solution flow: each arrow applies one index law.

Laws of Indices 28 (a4-028)

Difficulty: Foundation | Marks: 2

Evaluate: $\frac{7^3 \times 7^4}{7^6}$

Worked solution.

Step 1: Simplify the numerator by adding the exponents.

$$\frac{7^{3+4}}{7^6} = \frac{7^7}{7^6}$$

Both terms on top have the same base, so add the powers.

Step 2: Divide by subtracting exponents.

$$7^{7-6} = 7^1 = 7$$

$7 - 6 = 1$, and $7^1 = 7$.

Final answer: 7

Diagram.

$$\frac{7^3 \times 7^4}{7^6} \xrightarrow{\text{add (top)}} \frac{7^7}{7^6} \xrightarrow{\text{subtract}} 7^1 = 7$$

Solution flow: each arrow applies one index law.

Laws of Indices 29 (a4-029)

Difficulty: Foundation | Marks: 2

Evaluate: $(3^2)^3 \div (3^1)^4$

Worked solution.

Step 1: Apply the power-of-a-power law to both parts.

$$3^{2\times 3} \div 3^{1\times 4} = 3^6 \div 3^4$$

Multiply the exponents within each bracket.

Step 2: Subtract the exponents.

$$3^{6-4} = 3^2$$

$6 - 4 = 2$.

Step 3: Evaluate.

$$3^2 = 9$$

$3 \times 3 = 9$.

Final answer: 9

Diagram.

$$(3^2)^3 \div (3^1)^4 \xrightarrow{\text{powers of powers}} 3^6 \div 3^4 \xrightarrow{\text{subtract}} 3^2 = 9$$

Solution flow: each arrow applies one index law.

Laws of Indices 30 (a4-030)

Difficulty: Foundation | Marks: 3

Evaluate: $\frac{(2^{\frac{1}{2}})^6 \times (2^{-1})^4}{(2^{-4})^{-2}}$

Worked solution.

Step 1: Apply the power-of-a-power law to each bracket.

$$\frac{2^{\frac{1}{2}\times 6} \times 2^{-1\times 4}}{2^{-4\times(-2)}} = \frac{2^3 \times 2^{-4}}{2^8}$$

Multiply each pair of exponents: $\frac{1}{2} \times 6 = 3$, $-1 \times 4 = -4$, $-4 \times -2 = 8$.

Step 2: Simplify the numerator by adding exponents.

$$\frac{2^{3+(-4)}}{2^8} = \frac{2^{-1}}{2^8}$$

$3 + (-4) = -1$.

Step 3: Divide by subtracting the exponents.

$$2^{-1-8} = 2^{-9}$$

$-1 - 8 = -9$.

Step 4: Express as a fraction.

$$2^{-9} = \frac{1}{2^9} = \frac{1}{512}$$

$2^9 = 512$.

Final answer: $\dfrac{1}{512}$

Diagram.

$$\frac{(2^{\frac{1}{2}})^6 \times (2^{-1})^4}{(2^{-4})^{-2}} \xrightarrow{\text{powers of powers}} \frac{2^3 \times 2^{-4}}{2^8} \xrightarrow{\text{add (top)}} \frac{2^{-1}}{2^8} = 2^{-9} \xrightarrow{\text{subtract}} \frac{1}{512}$$

Solution flow: each arrow applies one index law.

Laws of Indices 31 (a4-031)

Difficulty: Foundation | Marks: 2

Simplify, leaving your answer as a single power of a: $\frac{a^5 \times a^3}{a^2}$

Worked solution.

Step 1: Simplify the numerator by adding the exponents.

$$\frac{a^{5+3}}{a^2} = \frac{a^8}{a^2}$$

Add the exponents on top: $5 + 3 = 8$.

Step 2: Divide by subtracting the exponents.

$$a^{8-2} = a^6$$

$8 - 2 = 6$.

Final answer: a^6

Diagram.

$$\frac{a^5 \times a^3}{a^2} \xrightarrow{\text{add (top)}} \frac{a^8}{a^2} \xrightarrow{\text{subtract}} a^6$$

Solution flow: each arrow applies one index law.

Laws of Indices 32 (a4-032)

Difficulty: Foundation | Marks: 2

Simplify: $\frac{c^4 d^{\frac{1}{2}}}{c^{-1} d^3}$

Worked solution.

Step 1: Apply the division law to each variable separately.

$$c^{4-(-1)} \times d^{\frac{1}{2}-3}$$

Subtract the exponent on the bottom from the exponent on the top for each base.

Step 2: Simplify each exponent.

$$c^5 d^{-\frac{5}{2}}$$

$4 - (-1) = 5$ and $\frac{1}{2} - 3 = -\frac{5}{2}$.

Final answer: $c^5 d^{-\frac{5}{2}}$

Diagram.

$$\frac{c^4 d^{\frac{1}{2}}}{c^{-1} d^3} \xrightarrow{\text{subtract powers per base}} c^{4-(-1)}\, d^{\frac{1}{2}-3} \xrightarrow{\text{simplify}} c^5 d^{-\frac{5}{2}}$$

Solution flow: each arrow applies one index law.

Laws of Indices 33 `(a4-033)`

Difficulty: Foundation | Marks: 2

Simplify: $\frac{12yz^{\frac{1}{3}}}{4yz^{\frac{1}{2}}}$

Worked solution.

Step 1: Divide the numerical coefficients and apply the division law to each variable.

$$\frac{12}{4} \times y^{1-1} \times z^{\frac{1}{3}-\frac{1}{2}}$$

Deal with the numbers, the y terms, and the z terms separately.

Step 2: Simplify each part.

$$3 \times y^0 \times z^{-\frac{1}{6}} = 3z^{-\frac{1}{6}}$$

$\frac{12}{4} = 3$, $y^0 = 1$, and $\frac{1}{3} - \frac{1}{2} = \frac{2-3}{6} = -\frac{1}{6}$.

Final answer: $3z^{-\frac{1}{6}}$

Diagram.

$$\frac{12yz^{\frac{1}{3}}}{4yz^{\frac{1}{2}}} \xrightarrow{\text{divide coeffs, subtract powers}} 3\, z^{\frac{1}{3}-\frac{1}{2}} \xrightarrow{\text{simplify}} 3z^{-\frac{1}{6}}$$

Solution flow: each arrow applies one index law.

Laws of Indices 34 `(a4-034)`

Difficulty: Foundation | Marks: 2

Simplify: $(ab^2)^3$

Worked solution.

Step 1: Raise each factor inside the bracket to the power 3.

$$a^3 \times (b^2)^3$$

When a product is raised to a power, each factor is raised to that power: $(ab)^n = a^n b^n$.

Step 2: Apply the power-of-a-power law to $(b^2)^3$.

$$a^3 b^6$$

$2 \times 3 = 6$.

Final answer: a^3b^6

Diagram.

$$\boxed{(ab)^n = a^nb^n}$$

$$(ab^2)^3 \xrightarrow[\text{raise each factor: } a^3(b^2)^3 = a^3b^6]{} = \ a^3b^6$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 35 (a4-035)

Difficulty: Foundation | Marks: 2

Simplify: $(mn^{\frac{1}{2}})^4$

Worked solution.

Step 1: Raise each factor inside the bracket to the power 4.

$$m^4 \times (n^{\frac{1}{2}})^4$$

Distribute the exponent across the product.

Step 2: Apply the power-of-a-power law.

$$m^4 n^{\frac{1}{2}\times 4} = m^4n^2$$

$\frac{1}{2} \times 4 = 2$.

Final answer: m^4n^2

Diagram.

$$\boxed{(ab)^n = a^nb^n}$$

$$(mn^{\frac{1}{2}})^4 \xrightarrow[\text{raise each factor: } m^4(n^{\frac{1}{2}})^4 = m^4n^2]{} = \ m^4n^2$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 36 (a4-036)

Difficulty: Foundation | Marks: 2

Simplify: $\frac{p^3q^4}{p^5q}$

Worked solution.

Step 1: Apply the division law to each variable separately.

$$p^{3-5} \times q^{4-1}$$

Subtract bottom exponent from top exponent for each base.

Step 2: Simplify.

$$p^{-2}q^{3}$$

$3-5=-2$ and $4-1=3$.

Final answer: $p^{-2}q^{3}$

Diagram.

$$\frac{p^3q^4}{p^5q} \xrightarrow{\text{subtract powers per base}} p^{3-5}\,q^{4-1} \xrightarrow{\text{simplify}} p^{-2}q^{3}$$

Solution flow: each arrow applies one index law.

Laws of Indices 37 (a4-037)

Difficulty: Foundation / Marks: 3

Evaluate: $(4^{-\frac{1}{2}})^2 \times (4^{-1})^{\frac{1}{2}}$

Worked solution.

Step 1: Apply the power-of-a-power law to each bracket.

$$4^{-\frac{1}{2}\times 2} \times 4^{-1\times\frac{1}{2}} = 4^{-1} \times 4^{-\frac{1}{2}}$$

$-\frac{1}{2}\times 2=-1$ and $-1\times\frac{1}{2}=-\frac{1}{2}$.

Step 2: Add the exponents since the bases are the same.

$$4^{-1+(-\frac{1}{2})} = 4^{-\frac{3}{2}}$$

$-1-\frac{1}{2}=-\frac{3}{2}$.

Step 3: Evaluate $4^{-\frac{3}{2}}$.

$$\frac{1}{4^{\frac{3}{2}}} = \frac{1}{(\sqrt{4})^3} = \frac{1}{2^3} = \frac{1}{8}$$

Take the reciprocal, then root-before-power: $\sqrt{4}=2$, then $2^3=8$.

Final answer: $\frac{1}{8}$

Diagram.

$$(4^{-\frac{1}{2}})^2 \times (4^{-1})^{\frac{1}{2}} \xrightarrow{\text{powers of powers}} 4^{-1}\times 4^{-\frac{1}{2}} \xrightarrow{\text{add powers}} 4^{-\frac{3}{2}} = \frac{1}{8}$$

Solution flow: each arrow applies one index law.

Laws of Indices 38 (a4-038)

Difficulty: Foundation | Marks: 2

Find the value of x: $4^x = \sqrt{4}$

Worked solution.

Step 1: Rewrite the right-hand side using index notation.

$$\sqrt{4} = 4^{\frac{1}{2}}$$

A square root is a power of $\frac{1}{2}$.

Step 2: Compare the exponents since the bases are equal.

$$4^x = 4^{\frac{1}{2}} \implies x = \frac{1}{2}$$

If $a^x = a^y$ then $x = y$.

Final answer: $x = \dfrac{1}{2}$

Diagram.

$$4^x = \sqrt{4} \xrightarrow{\text{same base}} 4^x = 4^{\frac{1}{2}} \xrightarrow{\text{equate powers}} x = \frac{1}{2}$$

Solution flow: each arrow applies one index law.

Laws of Indices 39 (a4-039)

Difficulty: Foundation | Marks: 3

Find the value of x: $9^x = \frac{1}{3}$

Worked solution.

Step 1: Write both sides with the same base. Since $9 = 3^2$, rewrite the left-hand side.

$$(3^2)^x = 3^{-1}$$

$\frac{1}{3} = 3^{-1}$ and $9 = 3^2$.

Step 2: Simplify the left-hand side using the power-of-a-power law.

$$3^{2x} = 3^{-1}$$

$(3^2)^x = 3^{2x}$.

Step 3: Since the bases are equal, set the exponents equal and solve.

$$2x = -1 \implies x = -\frac{1}{2}$$

Divide both sides by 2.

Final answer: $x = -\dfrac{1}{2}$

Diagram.

$$9^x = \frac{1}{3} \xrightarrow{\text{same base 3}} 3^{2x} = 3^{-1} \xrightarrow{\text{equate powers}} 2x = -1 \Rightarrow x = -\frac{1}{2}$$

Solution flow: each arrow applies one index law.

Laws of Indices 40 `(a4-040)`

Difficulty: Foundation | Marks: 2

Find the value of x: $\sqrt{5} \times 5^3 = 5^x$

Worked solution.

Step 1: Rewrite $\sqrt{5}$ as a power of 5.

$$5^{\frac{1}{2}} \times 5^3 = 5^x$$

$\sqrt{5} = 5^{\frac{1}{2}}$.

Step 2: Add the exponents on the left-hand side.

$$5^{\frac{1}{2}+3} = 5^{\frac{7}{2}} = 5^x$$

$\frac{1}{2} + 3 = \frac{1}{2} + \frac{6}{2} = \frac{7}{2}$.

Step 3: Compare exponents.

$$x = \frac{7}{2}$$

Both sides have base 5, so the exponents must be equal.

Final answer: $x = \dfrac{7}{2}$

Diagram.

$$\sqrt{5} \times 5^3 = 5^x \xrightarrow{\text{add powers}} 5^{\frac{1}{2}+3} = 5^x \xrightarrow{\text{equate powers}} x = \frac{7}{2}$$

Solution flow: each arrow applies one index law.

Laws of Indices 41 `(a4-041)`

Difficulty: Foundation | Marks: 1

Simplify, leaving your answer as a power: $10^6 \times 10^{-2}$

Worked solution.

Step 1: Add the exponents since the bases are the same.

$$10^{6+(-2)} = 10^{6-2}$$

The multiplication law says $a^m \times a^n = a^{m+n}$.

Step 2: Simplify the exponent.

$$10^4$$

$6 - 2 = 4$.

Final answer: 10^4

Diagram.

$$\boxed{a^m \times a^n = a^{m+n}}$$

$$10^6 \times 10^{-2} \xrightarrow[\text{add: } 6+(-2)=4]{} = 10^4$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 42 (a4-042)

Difficulty: Foundation | Marks: 1

Evaluate: $64^{\frac{1}{3}}$

Worked solution.

Step 1: Recognise that a power of $\frac{1}{3}$ means the cube root.

$$64^{\frac{1}{3}} = \sqrt[3]{64}$$

$a^{\frac{1}{3}} = \sqrt[3]{a}$.

Step 2: Evaluate the cube root.

$$4$$

$4 \times 4 \times 4 = 64$, so $\sqrt[3]{64} = 4$.

Final answer: 4

Diagram.

numerator $= 1$ (power)

$$64^{\frac{1}{3}} = \sqrt[3]{64} = 4$$

denominator $= 3$ (root)

Fractional index: the *denominator* gives the root, the *numerator* gives the power.

Laws of Indices 43 (a4-043)

Difficulty: Foundation | Marks: 2

Simplify, leaving your answer as a single power of w: $w^7 \div w^{-3}$

Worked solution.

Step 1: Subtract the exponents using the division law.

$$w^{7-(-3)}$$

$a^m \div a^n = a^{m-n}$. Be careful: subtracting a negative is the same as adding.

Step 2: Simplify the exponent.

$$w^{10}$$

$7 - (-3) = 7 + 3 = 10$.

Final answer: w^{10}

Diagram.

$$\boxed{a^m \div a^n = a^{m-n}}$$

$$w^7 \div w^{-3} \xrightarrow[\text{subtract: } 7-(-3)=10]{} = w^{10}$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 44 (a4-044)

Difficulty: Foundation | *Marks: 2*

Evaluate: $100^{\frac{3}{2}}$

Worked solution.

Step 1: Split the fractional exponent: square root first, then cube.

$$100^{\frac{3}{2}} = \left(100^{\frac{1}{2}}\right)^3$$

The denominator 2 means square root; the numerator 3 means cube.

Step 2: Take the square root of 100.

$$100^{\frac{1}{2}} = 10$$

$10 \times 10 = 100$.

Step 3: Cube the result.

$$10^3 = 1000$$

$10 \times 10 \times 10 = 1000$.

Final answer: 1000

Diagram.

numerator $= 3$ (power)

$$100^{\frac{3}{2}} = \left(\sqrt{100}\right)^3 = 10^3 = 1000$$

denominator $= 2$ (root)

Fractional index: the *denominator* gives the root, the *numerator* gives the power.

Laws of Indices 45 `(a4-045)`

Difficulty: Foundation | *Marks: 2*

Simplify: $\frac{(3^2)^4}{3^5}$

Worked solution.

Step 1: Apply the power-of-a-power law to the numerator.

$$\frac{3^{2\times 4}}{3^5} = \frac{3^8}{3^5}$$

Multiply the exponents: $2 \times 4 = 8$.

Step 2: Apply the division law.

$$3^{8-5} = 3^3$$

$8 - 5 = 3$.

Step 3: Evaluate.

$$3^3 = 27$$

$3 \times 3 \times 3 = 27$.

Final answer: 27

Diagram.

$$\frac{(3^2)^4}{3^5} \xrightarrow{\text{power of a power}} \frac{3^8}{3^5} \xrightarrow{\text{subtract}} 3^3 = 27$$

Solution flow: each arrow applies one index law.

Laws of Indices 46 `(a4-046)`

Difficulty: Foundation | *Marks: 1*

Evaluate: 2^{-3}

Worked solution.

Step 1: A negative exponent means take the reciprocal.

$$2^{-3} = \frac{1}{2^3}$$

$a^{-n} = \frac{1}{a^n}$.

Step 2: Evaluate 2^3.

$$\frac{1}{8}$$

$2 \times 2 \times 2 = 8$.

Final answer: $\frac{1}{8}$

Diagram.

$$\boxed{a^{-n} = \frac{1}{a^n}}$$

$$2^{-3} \xrightarrow[\text{negative power} \Rightarrow \text{reciprocal: } \frac{1}{2^3} = \frac{1}{8}]{} = \frac{1}{8}$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 47 `(a4-047)`

Difficulty: Foundation | Marks: 2

Simplify, leaving your answer as a single power of d: $d^{\frac{1}{2}} \times d^{\frac{3}{2}}$

Worked solution.

Step 1: Add the fractional exponents since the bases are the same.

$$d^{\frac{1}{2}+\frac{3}{2}}$$

The multiplication law says $a^m \times a^n = a^{m+n}$.

Step 2: Simplify the sum of the exponents.

$$d^{\frac{4}{2}} = d^2$$

$\frac{1}{2} + \frac{3}{2} = \frac{4}{2} = 2$.

Final answer: d^2

Diagram.

$$\boxed{a^m \times a^n = a^{m+n}}$$

$$d^{\frac{1}{2}} \times d^{\frac{3}{2}} \xrightarrow[\text{add: } \frac{1}{2} + \frac{3}{2} = 2]{} = d^2$$

Index-law card: the boxed general rule (teal) is applied to the expression.

Laws of Indices 48 `(a4-048)`

Difficulty: Foundation | Marks: 2

Find the value of x: $2^x = 16$

Worked solution.

Step 1: Rewrite 16 as a power of 2.

$$16 = 2^4$$

$2 \times 2 \times 2 \times 2 = 16$.

Step 2: Compare the exponents since the bases are equal.

$$2^x = 2^4 \implies x = 4$$

If $a^x = a^y$ then $x = y$.

Final answer: $x = 4$

Diagram.

$$2^x = 16 \xrightarrow{\text{same base 2}} 2^x = 2^4 \xrightarrow{\text{equate powers}} x = 4$$

Solution flow: each arrow applies one index law.

Laws of Indices 49 (a4-049)

Difficulty: Foundation | Marks: 2

Simplify: $\frac{5^4 \times 5^{-2}}{5^3}$

Worked solution.

Step 1: Simplify the numerator by adding the exponents.

$$\frac{5^{4+(-2)}}{5^3} = \frac{5^2}{5^3}$$

$4 + (-2) = 2$.

Step 2: Apply the division law.

$$5^{2-3} = 5^{-1}$$

$2 - 3 = -1$.

Step 3: Evaluate.

$$5^{-1} = \frac{1}{5}$$

A power of -1 means the reciprocal.

Final answer: $\frac{1}{5}$

Diagram.

$$\frac{5^4 \times 5^{-2}}{5^3} \xrightarrow{\text{add (top)}} \frac{5^2}{5^3} \xrightarrow{\text{subtract}} 5^{-1} = \frac{1}{5}$$

Solution flow: each arrow applies one index law.

Laws of Indices 50 (a4-050)

Difficulty: Foundation | Marks: 2

Evaluate: $8^{\frac{2}{3}}$

Worked solution.

Step 1: Split the fractional exponent: cube root first, then square.

$$8^{\frac{2}{3}} = \left(8^{\frac{1}{3}}\right)^2$$

The denominator 3 means take the cube root; the numerator 2 means square.

Step 2: Take the cube root of 8.

$$8^{\frac{1}{3}} = 2$$

$2 \times 2 \times 2 = 8$.

Step 3: Square the result.

$$2^2 = 4$$

$2 \times 2 = 4$.

Final answer: 4

Diagram.

numerator $= 2$ (power)

$$8^{\frac{2}{3}} \quad = \quad \left(\sqrt[3]{8}\right)^2 = 2^2 \quad = \; 4$$

denominator $= 3$ (root)

Fractional index: the *denominator* gives the root, the *numerator* gives the power.